

Chapter 9: Essential Probability Rules

School of Mathematical and Computational Sciences
University of Prince Edward Island

Ali Raisolsadat

Part II: Probability and Distributions

Randomness and Probability

When people hear the word “random,” they often think of something completely haphazard or patternless. In statistics, randomness has a more precise meaning.

Random phenomenon

A phenomenon is **random** when individual outcomes are uncertain, but a regular and predictable distribution emerges over many repetitions.

Probability

The probability of an outcome is the proportion of times that outcome would occur in a very long series of repetitions.

Example: Births

We cannot predict the sex of the next baby with certainty. However, when we look across a very large number of births, the proportions become remarkably stable. Probability describes this *long-run regularity*.

Two Ways to Determine Probability

How do we decide what probability to assign to an event? In practice, we usually rely on one of two approaches.

Theoretical probability

Derived from an underlying mathematical, physical, or biological model.

Example: A simplified genetic model suggests

$$P(\text{boy}) = 0.50.$$

The probability comes from the **model**, not from counting observations.

Empirical probability

Estimated from the long-run frequency observed in real data.

Example: Historical birth records may give

$$\widehat{P(\text{boy})} \approx 0.512.$$

The probability is estimated from **data**.

Both approaches can be used to construct a **probability model**, and both must obey the same mathematical probability rules.

The Frequentist Interpretation

Core idea

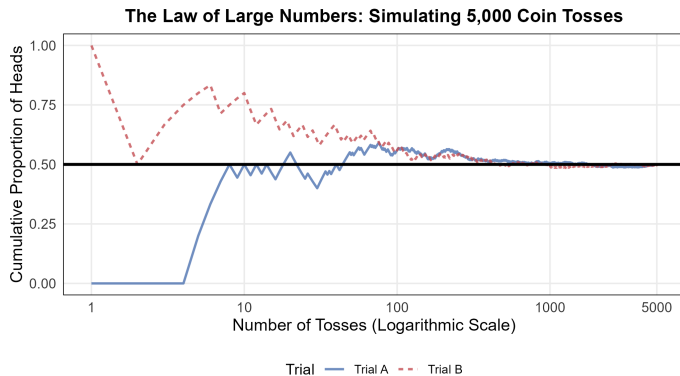
In the frequentist interpretation, probability is the **long-run relative frequency** of an outcome over many repeated trials under similar conditions.

- Individual trials remain uncertain.
- Small samples can vary substantially from one repetition to another.
- As the number of repetitions grows, relative frequencies tend to stabilize.
- A probability of 0.5 does **not** mean exactly half of the next ten trials must be successes.

Common cold example

Suppose a survey of 14,169 adults finds that 11.2% had a cold yesterday. A second random sample would not be expected to give exactly 11.2%, but many repeated samples would reveal a stable population pattern.

Long-Run Regularity: Coin Toss Simulation



- Each line is a separate sequence of 5,000 coin tosses.
- Early proportions can swing sharply because only a few tosses have occurred.
- As the number of tosses grows, both proportions move toward 0.5.
- The horizontal axis is logarithmic so the early variation is easier to see.

R Example: Simulating Coin Tosses

The simulation behind the previous figure uses a vector of 0s and 1s, where 1 represents heads.

```
set.seed(42)
n_tosses <- 5000

# Simulate two independent trials
tosses_A <- sample(c(0, 1), n_tosses, replace = TRUE)
tosses_B <- sample(c(0, 1), n_tosses, replace = TRUE)

# Running proportion of heads after each toss
prop_A <- cumsum(tosses_A) / (1:n_tosses)
prop_B <- cumsum(tosses_B) / (1:n_tosses)
```

What the R functions do

`sample()` generates the random tosses, while `cumsum()` keeps a running total of the number of heads.

The Law of Large Numbers

Law of Large Numbers

As the number of independent repetitions increases, the observed relative frequency tends to approach the underlying probability.

- Early results may fluctuate dramatically because each new observation has a large effect.
- With thousands of repetitions, one additional result has very little influence on the cumulative proportion.
- This long-run stabilization is one reason large samples are generally more dependable than very small samples.

Important caution

The Law of Large Numbers describes **long-run behaviour**. It does not force a short sequence to “balance out.” A coin has no memory of previous tosses.

The Illusion of Randomness

Which sequence is more likely when a fair coin is tossed six times?

Sequence A

H T H T T H

Sequence B

T T T H H H

The Illusion of Randomness

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Answer

Both specific sequences have probability

$$\left(\frac{1}{2}\right)^6 = \frac{1}{64}.$$

True randomness naturally produces streaks and clusters.

Personal Probability

Definition

A **personal probability** is a number between 0 and 1 that expresses an individual's scientifically informed judgement about how likely an outcome is.

Example

What is the probability that humans discover extraterrestrial life during this century?

This event cannot be repeated thousands of times. The probability must therefore be based on evidence, scientific models, and expert judgement rather than direct long-run repetition.

Regardless of its interpretation, every valid probability must obey the mathematical rules introduced later in this chapter.

Vocabulary of Probability Models

To use probability mathematically, we need to describe both the possible outcomes and how likely those outcomes are.

Sample space

The **sample space** S is the set of *all* possible outcomes of a random phenomenon.

Event

An **event** is one outcome or a collection of outcomes; mathematically, an event is a subset of S .

Probability model

A probability model specifies:

- 1 a sample space S ;
- 2 a rule for assigning probabilities to events.

For example, if we record only sex at birth, a simple sample space is $S = \{\text{Male}, \text{Female}\}$.

Example 9.5: Blood Types

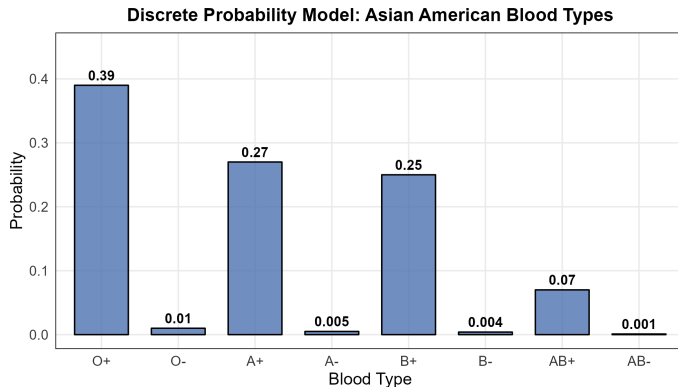
For the eight blood types,

$$S = \{O+, O-, A+, A-, B+, B-, AB+, AB-\}.$$

Type	$O+$	$O-$	$A+$	$A-$	$B+$	$B-$	$AB+$	$AB-$
Probability	.39	.01	.27	.005	.25	.004	.07	.001

- Each listed outcome is discrete.
- The probabilities are nonnegative and sum to 1.

Blood-Type Probability Model



The bars visualize the probability assigned to each discrete outcome.

- $O+$ has the largest probability in this model.
- Rare outcomes such as $AB-$ still belong to the sample space.
- Adding the eight probabilities gives exactly 1.

R Example: Representing a Discrete Probability Model

A probability model can be stored directly as vectors or as a data frame.

```
blood_data <- data.frame(  
  Type = factor(  
    c("O+", "O-", "A+", "A-", "B+", "B-", "AB+", "AB-"),  
    levels = c("O+", "O-", "A+", "A-", "B+", "B-", "AB+", "AB-")  
  ),  
  Probability = c(0.39, 0.01, 0.27, 0.005,  
                  0.25, 0.004, 0.07, 0.001)  
)  
  
sum(blood_data$Probability)  
# 1
```

Why check the sum?

A valid discrete probability model must assign probabilities that sum to 1.

Changing the Scope of a Model

Suppose we care only about the Rhesus factor.

$$S = \{Rh+, Rh-\}.$$

Rh factor	$Rh+$	$Rh-$
Probability	0.98	0.02

Key idea

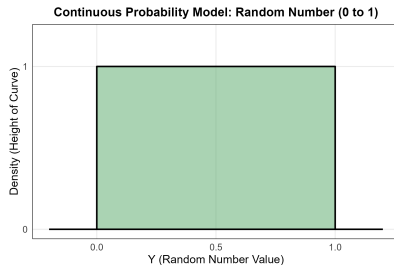
The sample space depends on how the random phenomenon is defined. A broader model can often be collapsed into a simpler model by combining outcomes.

Example 9.6: A Random Number from 0 to 1

Let Y be a random number selected from the interval $[0, 1]$.

$$S = \{y : 0 \leq y \leq 1\}.$$

- There are infinitely many possible decimal values.
- We cannot list every possible value and attach a positive probability to each one.
- Instead, probabilities are assigned to **intervals** as areas under a density curve.
- The entire area under the density curve must equal 1.



Discrete vs. Continuous Sample Spaces

Discrete

- Finite or countable outcomes
- Individual outcomes may have positive probability
- Probabilities can be listed in a table
- Example: blood type

Continuous

- An unbroken interval of values
- Individual points have probability 0
- Probabilities are areas under a curve
- Example: exact height

Four Fundamental Probability Rules

These rules must hold for **every** valid probability model, whether the sample space is discrete or continuous.

- 1 **Probabilities are proportions:**

$$0 \leq P(A) \leq 1.$$

A probability can never be negative or exceed 1.

- 2 **The entire sample space has probability 1:**

$$P(S) = 1.$$

- 3 **Add disjoint events:**

$$P(A \text{ or } B) = P(A) + P(B).$$

- 4 **Use the complement:**

$$P(A^c) = 1 - P(A).$$

Disjoint Events and the Addition Rule

Two events are **disjoint** when they have no outcomes in common.

$$P(A \text{ and } B) = 0.$$

Therefore,

$$P(A \text{ or } B) = P(A) + P(B).$$

For several pairwise disjoint events,

$$P(A \text{ or } B \text{ or } C) = P(A) + P(B) + P(C).$$

Do not double-count

This simple addition rule applies only when the events cannot occur together.

The Meaning of “OR” and “AND”

OR

Inclusive: at least one event occurs.

A alone, B alone, or both may occur.

AND

Intersection: both events occur at the same time.

This corresponds to the common outcomes of A and B .

When events overlap, the general addition rule is needed to correct for double-counting. That rule is introduced in Chapter 10.

Example 9.7: Positive Rhesus Factor

The positive blood-type outcomes are disjoint:

$$O+, \quad A+, \quad B+, \quad AB+.$$

Therefore,

$$\begin{aligned} P(Rh+) &= P(O+) + P(A+) + P(B+) + P(AB+) \\ &= 0.39 + 0.27 + 0.25 + 0.07 \\ &= \boxed{0.98}. \end{aligned}$$

Example 9.7: Complement Rule

The complement of having a positive Rhesus factor is having a negative Rhesus factor.

$$\begin{aligned}P(Rh-) &= 1 - P(Rh+) \\&= 1 - 0.98 \\&= \boxed{0.02}.\end{aligned}$$

Why complements are useful

It is often easier to subtract one probability from 1 than to add many separate probabilities.

Definition

A discrete probability model has a finite or countable list of outcomes. The probability of an event is the sum of the probabilities of the outcomes belonging to that event.

A valid discrete probability model must satisfy:

$$0 \leq p_i \leq 1 \quad \text{and} \quad \sum_i p_i = 1.$$

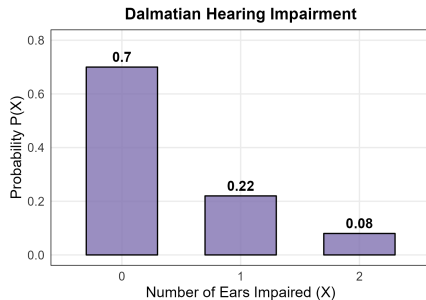
Example 9.8: Hearing Impairment in Dalmatians

A study of 5,333 Dalmatians recorded the number of ears with hearing impairment. Let X be the number of impaired ears for a randomly selected dog.

X	0	1	2
$P(X = x)$	0.70	0.22	0.08

The possible outcomes are the whole numbers 0, 1, and 2, so X is a **discrete random variable**. Notice that

$$0.70 + 0.22 + 0.08 = 1.$$



R Example: A Discrete Probability Distribution

The Dalmatian probability model can be represented with two short vectors.

```
# Possible values of X
x <- c(0, 1, 2)

# Corresponding probabilities
p_x <- c(0.70, 0.22, 0.08)

# Check that this is a valid probability model
sum(p_x)
# 1

# Probability of at least one impaired ear
sum(p_x[x >= 1])
# 0.30
```

Connection to the mathematics

For a discrete model, the probability of an event is found by **adding the probabilities of the outcomes inside the event.**

Some Hearing Impairment

“Some hearing impairment” means one or two impaired ears:

$$\begin{aligned}P(X \geq 1) &= P(X = 1) + P(X = 2) \\&= 0.22 + 0.08 \\&= \boxed{0.30}.\end{aligned}$$

Inequality signs matter

$$P(X \geq 1) = 0.30, \quad P(X > 1) = P(X = 2) = 0.08.$$

Continuous Probability Models and Density Curves

A continuous sample space contains an entire interval of possible values, so we cannot assign probability by listing individual points.

Density curve

A density curve:

- 1 lies on or above the horizontal axis;
- 2 has total area exactly equal to 1.

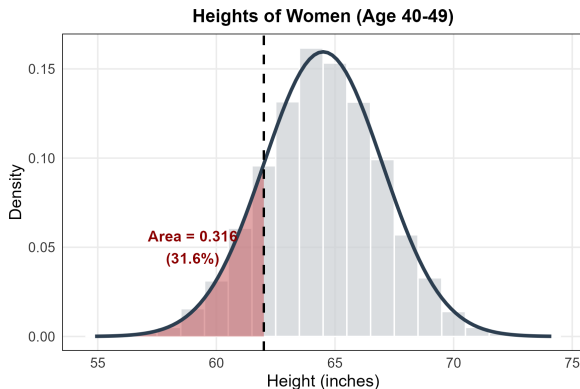
For a continuous random variable Y ,

$$P(a \leq Y \leq b) = \text{area under the density curve from } a \text{ to } b.$$

Important

Probability is represented by **area**, not by the height of the curve at one particular value.

Example 9.9–9.10: Women's Heights



The histogram shows simulated heights of women aged 40–49, while the smooth curve provides a continuous probability model.

- The total area under the smooth curve is 1.
- The shaded region to the left of 62 inches represents a probability.
- In this model, that area is approximately 0.316.

Probability as Area

For the model of women's heights shown previously,

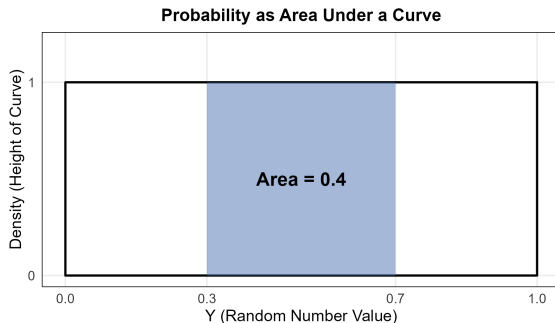
$$P(Y \leq 62) = 0.316.$$

- The probability is an **area**, not the height of the curve.
- The total area under the curve is 1.
- The shaded area represents 31.6% of the modeled population.

Example 9.11: Uniform Density

For a uniform random number between 0 and 1, the density has constant height 1. Therefore, probability is especially easy to compute:

$$\text{area} = \text{width} \times \text{height}.$$



$$\begin{aligned} P(0.3 \leq Y \leq 0.7) &= (0.7 - 0.3)(1) \\ &= \boxed{0.4}. \end{aligned}$$

The interval occupies 40% of the full width of $[0, 1]$, so it also contains 40% of the total probability.

R Example: Drawing a Uniform Probability Area

The shaded region in the uniform-density figure is produced by drawing one rectangle for the full density and another for the probability interval.

```
p_uniform_shaded <- ggplot() +  
  geom_rect(  
    aes(xmin = 0, xmax = 1, ymin = 0, ymax = 1),  
    fill = NA, color = "black"  
  ) +  
  geom_rect(  
    aes(xmin = 0.3, xmax = 0.7, ymin = 0, ymax = 1),  
    fill = "#4C72B0", alpha = 0.5  
  ) +  
  annotate("text", x = 0.5, y = 0.5,  
    label = "Area = 0.4")
```

The code visually connects the probability calculation $0.7 - 0.3 = 0.4$ with the corresponding area under the density curve.

Adding Disjoint Continuous Intervals

For $Y \sim \text{Uniform}(0, 1)$,

$$\begin{aligned}P(Y \leq 0.5 \text{ or } Y > 0.8) &= P(Y \leq 0.5) + P(Y > 0.8) \\&= 0.5 + 0.2 \\&= \boxed{0.7}.\end{aligned}$$

The intervals are disjoint, so their areas can be added directly.

The Probability of an Exact Value

For a continuous random variable,

$$P(Y = c) = 0$$

for every exact value c .

Why?

A single point has zero width, so the area above that point is zero.

Consequently,

$$P(Y \leq 0.8) = P(Y < 0.8).$$

Random Variables

Random phenomena often produce numerical outcomes. We use a random variable to give those outcomes a convenient mathematical name.

Definition

A **random variable** is a variable whose numerical value is determined by the outcome of a random phenomenon. We usually denote random variables by uppercase letters such as X or Y .

Probability distribution

The probability distribution of X specifies:

- which values X can take;
 - how probabilities are assigned to those values.
-
- X = number of impaired ears in a Dalmatian: $X \in \{0, 1, 2\}$.
 - Y = random number selected from $[0, 1]$: any real value in the interval is possible.

Two Types of Random Variables

Discrete random variable

Takes values from a finite or countable set.

Example:

$$X \in \{0, 1, 2\}.$$

Continuous random variable

Can take any value in an interval.

Example:

$$0 \leq Y \leq 1.$$

A measured variable can be intrinsically continuous even when recorded in rounded units.

Parameters vs. Statistics

Sample statistics

Computed from observed sample data:

$$\bar{x}, \quad s.$$

Distribution parameters

Describe the theoretical probability distribution:

$$\mu, \quad \sigma.$$

The mean of a random variable is also called its **expected value**.

Mean of a Discrete Random Variable

Let X have possible values $x_1, \dots, x_k$ with probabilities $p_1, \dots, p_k$.

Expected value

$$\mu = E(X) = \sum_{i=1}^k x_i p_i.$$

- The expected value is a **probability-weighted average**.
- Outcomes with larger probabilities contribute more strongly to μ .
- The value of μ does not need to be one of the possible individual outcomes.

Long-run interpretation

If the random experiment were repeated many times, the average observed value of X would tend toward μ .

Variance and Standard Deviation of a Random Variable

Once the mean μ is known, we can describe how much the random variable typically varies around that long-run average.

$$\sigma^2 = \sum_{i=1}^k (x_i - \mu)^2 p_i,$$

$$\sigma = \sqrt{\sigma^2}.$$

- Each squared distance $(x_i - \mu)^2$ is weighted by its probability.
- The variance σ^2 is measured in squared units.
- Taking the square root gives σ , which is expressed in the same units as X .

Connection to sample statistics

This is the theoretical analogue of the sample standard deviation s , but here the calculation uses the entire probability distribution.

Example 9.13: Expected Hearing Impairment

For $X \in \{0, 1, 2\}$ with probabilities 0.70, 0.22, 0.08,

$$\begin{aligned}\mu &= (0)(0.70) + (1)(0.22) + (2)(0.08) \\ &= \boxed{0.38}.\end{aligned}$$

Interpretation

The expected value need not be a possible individual outcome. A dog cannot have 0.38 impaired ears, but 0.38 is the long-run average across many dogs.

Example 9.13: Variance and Standard Deviation

$$\begin{aligned}\sigma^2 &= (0 - 0.38)^2(0.70) + (1 - 0.38)^2(0.22) \\ &\quad + (2 - 0.38)^2(0.08) \\ &= 0.10108 + 0.084568 + 0.209952 \\ &= \boxed{0.3956}, \\ \sigma &= \sqrt{0.3956} = \boxed{0.62897} \approx 0.63.\end{aligned}$$

Example 9.13: Computing μ and σ in R

The formulas translate directly into vectorized R operations.

```
# Define the outcomes (x) and their probabilities (p_x)
x <- c(0, 1, 2)
p_x <- c(0.70, 0.22, 0.08)

# 1. Calculate the Expected Value (mu)
mu <- sum(x * p_x)
print(mu)
# [1] 0.38

# 2. Calculate the Variance
variance <- sum((x - mu)^2 * p_x)

# 3. Calculate the Standard Deviation
sigma <- sqrt(variance)
print(sigma)
# [1] 0.6289674
```

R performs the multiplication element-by-element, so $x * p_x$ corresponds directly to the terms $x_i P(X = x_i)$ in the expected-value formula.

Probability, Risk, and Odds

In the life sciences, the same uncertain event may be communicated using **probability**, **risk**, or **odds**. These quantities are related, but they are not numerically identical.

If an event A occurs with probability p , then:

Risk

$$\text{Risk}(A) = p.$$

In health applications, risk usually refers to the probability of an undesirable outcome.

Odds

$$\text{Odds}(A) = \frac{p}{1 - p}.$$

Odds compare the probability that the event occurs with the probability that it does *not* occur.

Example 9.14: DVT as a Rare Event

Patients who remain immobilized for long periods can develop deep vein thrombosis (DVT). In the study described in the notes, 42 of 1,518 patients receiving Fragmin experienced DVT.

$$p = \frac{42}{1518} = 0.0277 \quad (2.77\%),$$

$$\text{Odds} = \frac{0.0277}{1 - 0.0277} = \frac{42}{1476} = 0.0285.$$

The odds can also be written approximately as

1 : 35.

How to read 1 : 35

For approximately every one patient who experiences DVT, about 35 do not.

Rare-event rule of thumb

When p is small, risk and odds are numerically similar because $1 - p \approx 1$.

Example 9.15: Sickle-Cell Anemia

If both parents carry the sickle-cell trait, the probability described in the notes that a child inherits the disease is

$$p = 0.25.$$

The **risk** is therefore 25%. The corresponding odds are

$$\begin{aligned}\text{Odds} &= \frac{0.25}{1 - 0.25} \\ &= \frac{0.25}{0.75} \\ &= \frac{1}{3} \approx 0.333.\end{aligned}$$

Thus the odds are 1 : 3: one occurrence for every three non-occurrences.

Important

For more common events, risk and odds can differ substantially. Always check which quantity a medical or scientific report is presenting.

Converting Odds Back to Probability

Starting from

$$\text{odds} = \frac{p}{1 - p},$$

solve for p :

$$p = \frac{\text{odds}}{1 + \text{odds}}.$$

For odds of $1/3$,

$$\begin{aligned} p &= \frac{1/3}{1 + 1/3} \\ &= \frac{1/3}{4/3} \\ &= \frac{1}{4} = 0.25. \end{aligned}$$

Summary of R Commands (Chapter 9)

The chapter uses a small set of R tools to connect probability formulas with simulation and computation.

R command	Role in this chapter
<code>sample(..., replace = TRUE)</code>	Simulates repeated random outcomes, such as coin tosses.
<code>cumsum(x)</code>	Creates a running total, useful for visualizing long-run relative frequency.
<code>data.frame(...)</code>	Stores outcomes and their probabilities together in a probability-model table.
<code>sum(x * p_x)</code>	Computes the expected value of a discrete random variable.
<code>sum((x - mu)^2 * p_x)</code>	Computes the probability-weighted variance.
<code>sqrt(variance)</code>	Converts variance into standard deviation.
<code>ggplot(), geom_bar(), geom_rect()</code>	Visualize discrete probabilities and continuous probability areas.

Key Takeaways

- Probability describes uncertainty using numbers from 0 to 1.
- Long-run relative frequency provides the frequentist interpretation.
- A probability model consists of a sample space and probability assignments.
- Disjoint events can be added; complements are found by subtracting from 1.
- Discrete probabilities are sums; continuous probabilities are areas.
- Random variables have theoretical means and standard deviations.
- Risk is a probability, while odds are $p/(1 - p)$.

Check Your Understanding

- 1 If $P(A) = 0.37$, what is $P(A^c)$?
- 2 If A and B are disjoint with probabilities 0.25 and 0.40, what is $P(A \text{ or } B)$?
- 3 For a continuous random variable, what is $P(X = 5)$?
- 4 If $p = 0.20$, what are the odds of the event?

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Answers

0.63; 0.65; 0; $0.20/0.80 = 0.25$, or 1 : 4.